

PMS Assignment-3

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Q2 Plumbing problem
long horizontal pipe of length L and Radius R
constant density and viscosity
fluid is at rest initially
at $t=0$, $\frac{dP}{dx} = \frac{P_0 - P_L}{L} = \gamma$

assumptions 1) velocity profile changes on x & t only

Writing Navier Stokes equation along z -direction

$$\rho \left(\frac{\partial v_z}{\partial t} + v_x \frac{\partial v_z}{\partial x} + v_\theta \frac{\partial v_z}{\partial \theta} + v_z \frac{\partial v_z}{\partial z} \right) = -\frac{\partial P}{\partial z} + \mu \left[\frac{1}{x} \frac{\partial}{\partial x} \left(x \frac{\partial v_z}{\partial x} \right) + \frac{1}{x^2} \frac{\partial^2 v_z}{\partial \theta^2} + \frac{\partial^2 v_z}{\partial z^2} \right] + \rho g_z$$

$$\rho \frac{\partial v_z}{\partial t} = \frac{P_L - P_0}{L} + \mu \left(\frac{1}{x} \frac{\partial}{\partial x} \left(x \frac{\partial v_z}{\partial x} \right) \right)$$

Boundary conditions for the problem

$$v(0, x) = 0$$

$$v(t, R) = v(t, 0) = 0$$

The above equation can't be solved analytically, so we will have to use numerical techniques

$$\text{Let } R = 0.5 \text{ m}, \rho = 1000 \text{ kg/m}^3, \mu = 1 \text{ Pa}\cdot\text{s}$$

using explicit Differencing scheme, the results can be seen in the attached code

For early times, we can assume

$u(r, t) = \left(1 - \frac{r^2}{R^2}\right) U(t)$, where $U(t)$ is the maximum velocity at
centreline of the pipe

$$\rho \frac{dU}{dt} = -\frac{4}{g} + \mu \frac{d^2 U}{dr^2}$$

$$\frac{dU}{dt} = -\frac{4}{g}$$

$$u(r, t) \approx \left(1 - \frac{r^2}{R^2}\right) \left(-\frac{4t}{g}\right)$$

The result can be compared with above one in code

Q3 Interphase Mixing - Paint and Emulsion Engineering
Using Fick's second law of Diffusion

$$\frac{\partial c}{\partial t} = D \frac{\partial^2 c}{\partial z^2}$$

$c \rightarrow$ concentration of the solute
 $t \rightarrow$ time
 $z \rightarrow$ spatial coordinate

For Phase I: $(-\infty < z < 0) \rightarrow c_I(-\infty, t) = c_I^0$

" Phase II: $(0 < z < \infty) \rightarrow c_{II}(\infty, t) = c_{II}^0$

Boundary condition at $z=0$ due to equilibrium at interface for $t > 0$

$$: c_{II}(0, t) = m c_I(0, t)$$

Since phases are immiscible, the flux at interface will be continuous

$$-D_I \frac{\partial c_I}{\partial z} \Big|_{z=0} = -D_{II} \frac{\partial c_{II}}{\partial z} \Big|_{z=0}$$

$$c(z, t) = c^0 \left(1 - \operatorname{erf} \left(\frac{z}{\sqrt{2Dt}} \right) \right), \text{ where } z \rightarrow (0, \infty)$$

$$\operatorname{erf}(z) = \frac{2}{\sqrt{\pi}} \int_0^z e^{-t^2} dt$$

The above equation is solution for semi-infinite solid with initial concentration c^0 and surface composition maintained at 0

For Phase I $\rightarrow$

$$c_I(z,t) = A + B \operatorname{erf}\left(\frac{z}{2\sqrt{D_I t}}\right)$$

For Phase II

$$c_{II}(z,t) = P + Q \operatorname{erf}\left(\frac{z}{2\sqrt{D_{II} t}}\right)$$

To find A, B, P, Q we will use initial conditions and boundary conditions

$$\text{For Phase I, at } z \rightarrow -\infty \quad c_I(z,t) = c_I$$

$$A - B = c_I$$

$$\text{At } z=0, \quad c_I = A$$

$$\text{For Phase II, at } z \rightarrow \infty, \quad c_{II} = c_{II}^0$$

$$P + Q = c_{II}^0$$

$$\text{At } z=0 \quad c_{II} = P = m c_I = m A$$

$$-D_I \left. \frac{\partial c_I}{\partial z} \right|_{z=0} = -D_{II} \left. \frac{\partial c_{II}}{\partial z} \right|_{z=0}$$

$$-D_I \times \frac{1}{2\sqrt{D_I t}} \times 1 \times B = -D_{II} \times \frac{1}{2\sqrt{D_{II} t}} \times 1 \times Q$$

$$\sqrt{D_I} B = \sqrt{D_{II}} Q$$

$$P + Q = c_{II}^0 \rightarrow mA + \sqrt{\frac{D_I}{D_{II}}} B = c_{II}^0$$

$$A - B = c_I^0 \rightarrow mA - mB = m c_I^0$$

$$B \left[m + \sqrt{\frac{D_I}{D_{II}}} \right] = C_{II}^0 - m C_I^0$$

$$\text{let } \sqrt{\frac{D_I}{D_{II}}} = \phi, \quad B = \frac{C_{II}^0 - m C_I^0}{(m + \phi)}$$

$$A = C_I^0 + B = \frac{C_{II}^0 + \phi C_I^0}{(m + \phi)}$$

$$P = \frac{m}{m + \phi} \left[C_{II}^0 + \phi C_I^0 \right]$$

$$Q = \phi B = \frac{\phi}{m + \phi} \left[C_{II}^0 - m C_I^0 \right]$$

$$\text{Phase I: } C_I(z, t) = \frac{1}{m + \phi} \left[C_{II}^0 + \phi C_I^0 \right] + \frac{1}{m + \phi} \left[C_{II}^0 - m C_I^0 \right] \times \text{erf} \left(\frac{z}{\sqrt{4 D_{II} t}} \right)$$

$$\text{Phase II: } C_{II}(z, t) = \frac{m}{m + \phi} \left[C_{II}^0 + \phi C_I^0 \right] + \frac{\phi}{m + \phi} \left[C_{II}^0 - m C_I^0 \right] \times \text{erf} \left(\frac{z}{\sqrt{4 D_{II} t}} \right)$$

Q4 Catalyst deactivation in continuous reactor

$$R = k C^2 A$$

$$\frac{dq}{dt} = -k_1 C_A \rightarrow \text{catalyst deactivation rate}$$

Non-dimensionalising the equations using $\varepsilon = \frac{k_1}{k} \ll 1$

$$\tilde{c} = \frac{c}{c_0}, \quad \tilde{a} = \frac{a}{a_0}$$

$$\tilde{t} = k t c_0$$

where c_0 & a_0 are the initial concentration and catalyst activity

Assuming $a_0 = 1$

$$\frac{d\tilde{c}}{d\tilde{t}} = -\tilde{c}^2 \tilde{a} \quad \& \quad \frac{d\tilde{a}}{d\tilde{t}} = -\varepsilon \tilde{c} \tilde{a}$$

Inner solution $\rightarrow$ for inner solution we will solve concentration's differential equation on reaction time scale, assuming $\tilde{a}$ is approx 1

$$\frac{d\tilde{c}}{d\tilde{t}} = -\tilde{c}^2 \Rightarrow \int \frac{d\tilde{c}}{\tilde{c}^2} = \int -1$$

$$\frac{-1}{\tilde{c}} = -\tilde{t} + A$$

$$\text{At } \tilde{t}=0, \tilde{c}=1$$

$$A = -1 \quad \tilde{c}(\tilde{t}) = \frac{1}{1+\tilde{t}}$$

$$C(t) = \frac{C_0}{1+k_t C_0 t}$$

Outer solution: We consider catalyst deactivation on slower timescale $\tilde{\tau} = \varepsilon \tilde{t}$
 assuming $\tilde{c} \approx 1$ [as concentration changes quickly to ss]

We get

$$\frac{d\tilde{a}}{d\tilde{\tau}} = -\tilde{a} \quad \frac{d\tilde{a}}{\tilde{a}} = \int -d\tilde{\tau}$$

$$\ln \tilde{a} = -\tilde{\tau} + B$$

$$\text{At } \tilde{\tau}=0, \tilde{a}=0$$

$$B=0$$

$$\tilde{a} = e^{-\tilde{\tau}}$$

$$a = e^{-k_e t}$$

$$\tilde{c} = \tilde{c}_0 + \varepsilon \tilde{c}_1 + \dots$$

$$\tilde{a} = \tilde{a}_0 + \varepsilon \tilde{a}_1 + \dots$$

$$\tilde{c} = \frac{1}{1+\tilde{\tau}} \quad \& \quad \tilde{a}_0 = e^{-\tilde{\tau}}$$

$$\frac{d\tilde{c}}{d\tilde{\tau}} = -\tilde{c}^2 \tilde{a}$$

$$\frac{d}{d\tilde{\tau}} [\tilde{c}_0 + \varepsilon \tilde{c}_1] = -[\tilde{c}_0 + \varepsilon \tilde{c}_1]^2 [\tilde{a}_0 + \varepsilon \tilde{a}_1]$$

$$\frac{d\tilde{c}_0}{d\tilde{\tau}} + \varepsilon \frac{d\tilde{c}_1}{d\tilde{\tau}} = (-\tilde{c}_0 \tilde{a}_0 - \varepsilon \tilde{c}_1 \tilde{a}_0 - \varepsilon \tilde{c}_0 \tilde{a}_1 - \varepsilon^2 \tilde{c}_1 \tilde{a}_1) (\tilde{c}_0 + \varepsilon \tilde{c}_1)$$

$$\frac{d\tilde{c}_1}{d\tilde{\tau}} = -2\tilde{c}_0 \tilde{c}_1 \tilde{a}_0 - \tilde{c}_0^2 \tilde{a}_1$$

$$\frac{d\tilde{c}_1}{d\tilde{t}} = \frac{-2e^{-\tilde{t}}\tilde{c}_1}{1+\tilde{t}} - \frac{1}{(1+\tilde{t})^2}\tilde{a}_1$$

$$\frac{d\tilde{a}_1}{d\tilde{t}} = -\tilde{c}_0\tilde{a}_1 - \tilde{c}_1\tilde{a}_0$$

$$\frac{d\tilde{a}_1}{d\tilde{t}} = \frac{-\tilde{a}_1}{(1+\tilde{t})} - \tilde{c}_1 e^{-\tilde{t}}$$

can only be solved numerically

Q5 Bubble dynamics: combustion engineering

a) Material balance equation

$$\frac{d}{dt} \left(\frac{4}{3} \pi R^3 C \right) = -k_c 4 \pi R^2 C^*$$

$$P = C R^* T \quad [\text{ideal gas law}]$$

$$C = \frac{P}{R^* T}$$

$$\frac{d}{dt} \left[\frac{4}{3} \pi R^3 \frac{P}{R^* T} \right] = -k_c 4 \pi R^2 C^*$$

$$\frac{4}{3} \pi \frac{P}{R^* T} \frac{d}{dt} (R^3) + \frac{4}{3} \pi \frac{R^3}{R^* T} \frac{dP}{dt} = -k_c 4 \pi R^2 C^*$$

$$\frac{dz}{dt} = \beta R \Rightarrow \frac{dP}{dt} = S_g \beta R$$

$$\frac{P}{R^* T} \frac{dR}{dt} + \frac{S_g \beta R^2}{3 R^* T} = -k_c C^*$$

$$\frac{dR}{dt} = -k_c \frac{C^* R^* T}{P} - \frac{S_g \beta R^2}{3 P}$$

b) Let $k_c C^* R^* T = a$ and $\frac{S_g \beta}{3} = b$

$$\frac{dR}{dt} = -\frac{a}{P} - b R^2$$

$$\frac{dR}{dt} + b R^2 = -\frac{a}{P}$$

$$p = \frac{-a}{\frac{dR}{dt} + bR^2}$$

$$\frac{dp}{dt} = \left(\frac{a}{\frac{dR}{dt} + bR^2} \right)^2 \times \left(\frac{d^2R}{dt^2} + 2bR \right)$$

$$3bR \left(\left(\frac{dR}{dt} \right)^2 + b^2R^4 + 2bR^2 \frac{dR}{dt} \right) = a \frac{d^2R}{dt^2} + 2abR$$

$$a \frac{d^2R}{dt^2} - 3bR \left(\frac{dR}{dt} \right)^2 - b^2R^3 \frac{dR}{dt} + 2abR - 3b^3R^5 = 0$$

The above equation can not be solved analytically, so using for solving it numerically

Q6 Solidification dynamics (metallurgical system)

1D case.

It is a semi infinite solid, which means it can be assumed that the solid extends infinitely in +ve-x-direction

Heat equation:

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2}$$

Boundary conditions:

$$T(x, 0) = T_f \text{ for } x > 0$$

$$T(0, t) = T_0 \text{ for } t > 0$$

using similarity variable n , used in semi-infinite domains

$$n = \frac{x}{2\sqrt{\alpha t}}$$

Solution of temperature profile is expressed using erf as

$$T(n) = T_0 + (T_f - T_0) \operatorname{erf}(n)$$

$$T(x, t) = T_0 + (T_f - T_0) \operatorname{erf} \left(\frac{x}{2\sqrt{\alpha t}} \right)$$

2D case:

Heat equation for 2D:

$$\frac{\partial T}{\partial t} = \kappa \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right)$$

BC $\rightarrow T(x, y, 0) = T_b$ for all x & y
 $T(0, 0, t) = T_o$ for $t > 0$

For semi infinite domain:

$$\frac{T(x, y, t) - T_o}{T_b - T_o} = \left(\frac{T(x, t) - T_o}{T_b - T_o} \right) \kappa \left(\frac{T(y, t) - T_o}{T_b - T_o} \right)$$

$$\frac{T(x, y, t) - T_o}{T_b - T_o} = \text{erf} \left(\frac{x}{2\sqrt{\kappa t}} \right) \text{erf} \left(\frac{y}{2\sqrt{\kappa t}} \right)$$

so, $T(x, y, t) = T_o + (T_b - T_o) \text{erf} \left(\frac{x}{2\sqrt{\kappa t}} \right) \text{erf} \left(\frac{y}{2\sqrt{\kappa t}} \right)$

Q1 Microelectronics processing

Considering just mass transfer of a single reactant and assuming that reactor operates in steady state. Also assuming that reactant gases is dilute and flow is laminar, fully developed and diffusion is primary mode of mass transfer. Ignoring heat transfer in simplified model.

Concentration of reactant gas depends on x & z within annulus b/w the wafer and reactor wall.

so, governing equation is,

$$\frac{1}{r} \left(\frac{\partial}{\partial r} \left(r D \frac{\partial C}{\partial r} \right) \right) + \frac{\partial}{\partial z} \left(D \frac{\partial C}{\partial z} \right) = 0$$

Boundary condition:

At reactor wall $r = R$ $\frac{\partial C}{\partial r} = 0$

At $r = 0$ $\frac{\partial C}{\partial r} = 0$

At $z = 0$ (gas inlet) $C(x, 0) = C_o$

at $z = L$ (yes outlet) $\frac{\partial C}{\partial z} = 0$

separation of variables $\rightarrow C(x, z) = R(x)Z(z)$

$$\frac{1}{xR} \frac{d}{dx} \left(x \frac{dR}{dx} \right) Z + \frac{1}{Z} \frac{d}{dz} \left(\frac{dZ}{dz} \right) = 0$$

$$\frac{1}{xR} \frac{d}{dx} \left(x \frac{dR}{dx} \right) = -\frac{1}{Z} \frac{d^2 Z}{dz^2} = \lambda^2$$

For Z ,

$$\frac{d^2 Z}{dz^2} + \lambda^2 Z = 0$$

$$Z(z) = A \sin(\lambda z) + B \cos(\lambda z)$$

$$\frac{dZ}{dz} = A \lambda \cos(\lambda z) - B \sin(\lambda z)$$

$$B = A \cot(\lambda L)$$

$$Z(z) = \frac{A}{\sin(\lambda L)} \left[\sin(\lambda z) \sin(\lambda L) + \cos(\lambda z) \cos(\lambda L) \right]$$

$$Z(z) = A \operatorname{cosec}(\lambda L) \cos(\lambda(L-z))$$

Now for R ,

$$\frac{1}{x} \frac{d}{dx} \left(x \frac{dR}{dx} \right) + \lambda^2 R = 0$$

It is a form of Bessel equation,

$$\text{so, } R(x) = C J_0(\lambda x) + D Y_0(\lambda x)$$

for $x=0$ $R \rightarrow$ finite, so $D=0$

$$R(x) = C J_0(\lambda x)$$

$$\frac{dR}{dx} = -C \lambda J_1(\lambda x) = 0 \quad \text{at } x=R$$

$$\Rightarrow J_1(\lambda R) = 0$$

$$C(x, z) = \sum_{m=1}^{\infty} A^* J_0(\lambda_m x) \operatorname{cosec}(\lambda_m L) \cos(\lambda_m(L-z))$$

At $z=0$ $c=c_0$

$$c_0 = \sum_{n=1}^{\infty} A^n J_0(\lambda_n x) \cos(\lambda_n h) \cos(\lambda_n h)$$

$$\int_0^R c_0 J_0(\lambda_n x) x dx = A^n \cos(\lambda_n h) \int_0^R J_0^2(\lambda_n x) x dx$$

$$A^n = \frac{c_0 \left(\int_0^R J_0^2(\lambda_n x) x dx \right) J_0(\lambda_n h)}{\int_0^R J_0^2(\lambda_n x) x dx}$$

Q7 skin patch for drug release

assuming initial concentration of c_0 in top layer

let gel thickness be L

$$\frac{\partial c}{\partial t} = D \frac{\partial^2 c}{\partial y^2} \rightarrow \text{Diffusion along } y\text{-direction}$$

BCs

$$c(y, 0) = c_0 \text{ for } 0 < x < L$$

$$c(0, t) = 0 \text{ for } t > 0$$

$$\text{At } x=L \quad -D \left. \frac{\partial c}{\partial y} \right|_{y=L} = k C(L, t)$$

$k \rightarrow$ rate at which wound consumes active ingredient

$$C(y, t) = Y(y) T(t)$$

$$Y \frac{dT}{dt} = T \frac{d^2 Y}{dy^2}$$

$$\frac{1}{T} \frac{dT}{dt} = \frac{1}{Y} \frac{d^2 Y}{dy^2} = -\alpha^2$$

$$\text{For } Y: \quad \frac{d^2 Y}{dy^2} + \alpha^2 Y = 0$$

$$Y(y) = A \sin(\alpha y) + B \cos(\alpha y)$$

$$\text{At } y=0 \quad Y=0 \rightarrow B=0$$

$$\psi(y) = A \sin(\alpha y)$$

$$\text{At } y=L \rightarrow -D \frac{\partial \psi}{\partial y} = kL \Rightarrow -D \alpha A \cos(\alpha L) = kA \sin(\alpha L)$$

$$\alpha \cot(\alpha L) = -\frac{k}{D}$$

$$\psi_m(y) = A \sin(\alpha_m y)$$

$$\text{For } T: \frac{dT}{dt} + \alpha^2 T = 0 \Rightarrow T_m = C e^{-\alpha_m^2 t}$$

$$C(y, t) = \sum_{n=1}^{\infty} A^* \sin(\alpha_n y) e^{-\alpha_n^2 t}$$

$$\text{At } t=0, C=C_0$$

$$\Rightarrow C_0 = \sum_{n=1}^{\infty} A^* \sin(\alpha_n y)$$

$$\int_0^L C_0 \sin(\alpha_n y) dy = A^* \int_0^L \sin^2(\alpha_n y) dy$$

$$A^* = \frac{C_0 \int_0^L \sin(\alpha_n y) dy}{\int_0^L \sin^2(\alpha_n y) dy} = C_0^*$$

Flux at time t and location y :

$$J = -D \frac{\partial C}{\partial y} = -C_0^* D \sum_{n=1}^{\infty} \alpha_n \cos(\alpha_n y) e^{-\alpha_n^2 t}$$

$$C_{eff} = 0.005 C_0$$

$$h_{eff} = 0.995 L$$

given amount of active ingredient $M \rightarrow$

$$M = \int_0^T J A dt = -A \int_0^T C_0^* D \sum_{n=1}^{\infty} \alpha_n \cos(\alpha_n y) e^{-\alpha_n^2 t} dt$$

$$M = A D C_0^* \sum_{n=1}^{\infty} \alpha_n \cos(\alpha_n y) \frac{1}{\alpha^2} e^{-\alpha_n^2 t} \Big|_0^T$$

$$M = AD C_0 \sum_{n=1}^{\infty} \frac{1}{\kappa_n} \cos(\alpha_n y) [1 - e^{-\kappa_n^2 T}]$$

From this T can be obtained